01 / Complete case and authoring reference

Leg Pain on Walking

CLINICAL CHALLENGE 01 | CURRENT SOURCE SNAPSHOT | 18 OCTOBER 2026

This guide captures the current learner-facing case, its conditional debrief, and all recognized answer phrases. Use the reference IDs to request edits or specify content for another case using the same game functionality.

P01 | Patient presentation

A 54-year-old man reports gradually worsening pain in both legs over the past year. He has begun cutting back on walks and errands.

G01 | Shared game behaviour

Explore History, Examination and Imaging tabs without an information budget. Each inquiry can be requested once. Ask at least one history question to unlock examination inquiries.

New findings appear at the top of the Patient Record, with the initial referral at the bottom. Entries show their category, question label and response, with a small category icon.

Optionally enter up to three ranked working diagnoses. Reorder by dragging or by using the arrows. Remove and replace entries while the case is active. No revision counter or numerical score is shown.

Submit a free text final diagnosis and a required confidence rating from 1 to 5. There are two diagnostic attempts. After an incorrect first answer, further inquiry and differential revision remain available.

A correct answer or a second incorrect answer ends the case and reveals the debrief. Restart clears the record, differential, confidence and attempts. Progress is not saved between page loads.

The desktop workspace places Clinical Inquiry above Working Differential on the left and Patient Record above Final Diagnosis on the right. It stacks on smaller screens. Inquiry has a fixed height; the other cards grow to a shared cap and then scroll. The cap fits Final Diagnosis with the longer retry message.

G02 | Guide contents

02-04 Inquiry options and responses 05-08 Debrief content and conditional findings 09-10 Confidence feedback and submission messages 11-12 Complete phrase lists 13 Shared interface wording 14 Brief for a new case

02 / History inquiries - first four

Burgundy bands show the actual button label, also used in the Patient Record. The response beneath is exact case text. Stored full questions are included in grey where the shorter label differs.

H01 | History

What triggers the pain?

Stored full question: What brings on the pain?

Walking or standing for several minutes brings on aching and heaviness in his legs.

H02 | History

What relieves it?

Stored full question: What makes it better?

Sitting usually relieves symptoms within a few minutes. Simply stopping while standing helps much less.

H03 | History

Does leaning forward help?

Yes. Leaning over a shopping cart lets him walk much farther.

H04 | History

Where does the pain spread?

Stored full question: Where does the discomfort travel?

It involves both buttocks and the back of the thighs, sometimes extending into the calves.

NOTES FOR YOUR NEXT CASE / REQUESTED CHANGES

03 / History inquiries - remaining four

H05 | History

Does slope affect walking?

Stored full question: Does walking uphill or downhill differ?

Downhill walking is more difficult than uphill walking.

H06 | History

Can he cycle?

Stored full question: Can he ride a bicycle?

He tolerates a stationary bicycle much better than upright walking.

H07 | History

Any vascular risk factors?

He has hypertension and previously smoked; he has no known peripheral arterial disease.

H08 | History

Any neurological red flags?

Stored full question: Any new bladder, saddle, or progressive weakness symptoms?

No bowel or bladder changes, saddle sensory changes, or progressive focal weakness.

NOTES FOR YOUR NEXT CASE / REQUESTED CHANGES

04 / Examination and imaging inquiries

Examination requires at least one history inquiry. This case has one Imaging option and no previous lumbar MRI or CT result.

E01 | Examination

How are the leg pulses?

Both feet are warm, and distal pulses are palpable bilaterally.

E02 | Examination

Neurological exam findings?

Stored full question: What is the neurological examination at rest?

Lower-extremity strength, reflexes, and light-touch sensation are broadly normal at rest.

E03 | Examination

Straight leg raise result?

Stored full question: What does straight leg raise show?

Straight leg raise does not reproduce his characteristic symptoms.

E04 | Examination

Lumbar movement findings?

Stored full question: Does lumbar movement provoke symptoms?

Extension is restricted and increases discomfort; flexion is better tolerated.

I01 | Imaging

Previous lumbar imaging?

Stored full question: Any previous lumbar MRI or CT?

He has not had a lumbar MRI or CT.

NOTES FOR YOUR NEXT CASE / REQUESTED CHANGES

05 / Debrief - diagnosis and learner summary

The debrief appears only when the case ends. Its section order is Diagnosis, Important alternatives, Diagnostically useful findings, Next steps, then Further reading.

D00 | Main title

Debrief

D01 | Diagnosis

Lumbar spinal stenosis causing neurogenic claudication

The patient's bilateral leg discomfort is brought on by standing or walking and relieved by sitting or lumbar flexion. His improvement when leaning forward, along with greater difficulty walking downhill than uphill, makes neurogenic claudication particularly likely. This remains a clinical working diagnosis, as lumbar imaging has not yet confirmed stenosis.

D02 | Learner summary

Your diagnosis [learner's final submitted wording]

Confidence [Low / Moderate / High] ([rating]/5)

Bracketed fields are learner-specific placeholders in this guide. The game shows actual submitted text and the confidence rating from the final attempt. A short conditional observation follows, as documented in sections 09-10.

D03 | Your ranked differential

Shown only when the learner entered a working differential. The original wording and final ranking are preserved. Each entry receives one compact label and a small icon.

D03 | Leading diagnosis

A recognized term for the intended diagnosis. Subtle positive check icon.

D03 | Relevant alternative

A recognized diagnosis in one of this case's alternative families. Subtle positive check icon.

D03 | Not assessed

No match in this case's predefined lists. Neutral information icon.

When at least one entry is unrecognized, the game displays:

Not assessed means there is no predefined feedback for this diagnosis. It does not mean incorrect or inappropriate.

NOTES FOR YOUR NEXT CASE / REQUESTED CHANGES

06 / Debrief - alternatives and next steps

D04 | Important alternatives

D04-1 | Peripheral arterial disease

Exertional leg discomfort and his vascular risk factors make peripheral arterial disease an important alternative. Vascular claudication typically improves with rest even without sitting or bending forward, whereas this patient's symptoms are strongly influenced by posture. Palpable pulses do not exclude arterial disease.

D04-2 | Lumbar radicular pain

Radicular pain can radiate into the legs, but it more often follows a recognizable nerve root distribution and may be accompanied by neurological deficits or positive neural tension testing. These findings are not prominent in this presentation.

D04-3 | Peripheral neuropathy

Peripheral neuropathy is another possible cause of leg discomfort, particularly when sensory symptoms are present. However, symptoms are generally more persistent and distally distributed rather than consistently provoked by standing and relieved by spinal flexion.

D05 | Next steps

D05-1 | Complete the assessment

Clarify the effect on walking tolerance and daily function, complete the neurological and vascular examinations, and screen for new neurological red flags.

D05-2 | Investigations

Consider lumbar MRI when symptoms remain functionally limiting or when imaging would guide further management, particularly before an intervention. If peripheral arterial disease remains a concern, obtain an ankle brachial index.

D05-3 | Management

Discuss individualized conservative management, including physiotherapy, exercise, and strategies to maintain activity. Persistent functional limitations may warrant specialist assessment. New bowel or bladder dysfunction, saddle sensory changes, or progressive neurological weakness require urgent evaluation.

Each alternative and next step is a separate restrained box. A future case can also provide a Must not miss subsection for dangerous alternatives; this case does not have one.

07 / Debrief - primary useful findings

All three primary findings appear. Each has an Asked or Not asked badge according to the inquiry IDs selected. Asked is green; Not asked is grey. The text explains significance and does not repeat the full patient response.

D06-H01 | Symptoms provoked by standing or walking

Asked

Neurogenic and vascular claudication can both produce exertional leg symptoms. The relationship with prolonged standing provides an additional clue toward a spinal mechanism.

Not asked

Asking whether standing as well as walking provokes symptoms could help distinguish a spinal mechanism from exertional leg pain alone.

D06-H02 | Relief with sitting rather than standing rest

Asked

Relief with sitting is especially informative because it suggests that changing spinal posture matters, rather than simply stopping physical activity.

Not asked

Comparing relief from sitting with relief from simply stopping while standing could help establish whether spinal posture influences the symptoms.

D06-H03 | Improvement with lumbar flexion

Asked

The shopping cart sign supports neurogenic claudication because lumbar flexion can increase the available space around neural structures and reduce symptom provocation.

Not asked

Asking whether leaning forward improves walking tolerance could help establish whether lumbar flexion relieves the symptoms.

NOTES FOR YOUR NEXT CASE / REQUESTED CHANGES

08 / Debrief - supporting findings and reading

At most one additional supporting finding appears, and only if it was asked. Priority is H05, then H06, then E04. If none was asked, no supporting entry appears. There is no separate missed-inquiries list.

D07-H05 | Walking slope | Asked only

Greater difficulty walking downhill than uphill adds support for a spinal mechanism because downhill walking tends to involve more lumbar extension.

D07-H06 | Cycling tolerance | Asked only

Better tolerance of cycling than upright walking adds support for a posture dependent pattern rather than exertion alone.

D07-E04 | Lumbar movement | Asked only

Discomfort with extension and better tolerance of flexion reinforce the relationship between symptoms and spinal posture.

D08 | Further reading

Both optional readings appear in a compact section. Each opens in a new tab.

D08-1 | Lumbar Spinal Stenosis: Diagnosis and Management

American Family Physician, 2024

A practical clinical review covering diagnosis, investigations, and management of lumbar spinal stenosis.

https://www.aafp.org/afp/2024/0400/lumbar-spinal-stenosis

D08-2 | 2024 Guideline for the Management of Lower Extremity Peripheral Artery Disease

American Heart Association

A concise guideline summary reviewing the assessment of peripheral arterial disease, including the role of the ankle brachial index.

https://professional.heart.org/en/science-news/2024-guideline-for-the-management-of-lower-extremity-peripheral-artery-disease/top-things-to-know

NOTES FOR YOUR NEXT CASE / REQUESTED CHANGES

09 / Confidence feedback - correct final diagnosis

Low confidence means 1-2, Moderate means 3, and High means 4-5. The game uses the final submitted answer and rating. These paragraphs appear inside the learner summary, not in a separate large section.

C-CORRECT-HIGH | High confidence

Well done. You identified the most likely diagnosis with high confidence. The symptom pattern you explored provides strong support for your answer.

C-CORRECT-MODERATE | Moderate confidence

Well done. You identified the most likely diagnosis with moderate confidence. The findings below can help you feel more assured about this impression, particularly the relationship between symptoms and spinal posture.

C-CORRECT-LOW | Low confidence

Well done. You reached the most likely diagnosis even though you felt uncertain. Read through the key findings below, especially relief with sitting and forward flexion, to see why your answer fits and build confidence for similar presentations.

C-CORRECT-HIGH-LIMITED | High confidence without the key inquiry combination

Well done. You identified the most likely diagnosis with high confidence. The findings below explain why your answer fits this presentation.

C-EVIDENCE | Conditional additional sentence

The basic evidence threshold is H01 plus either H02 or H03. If met, no extra evidence sentence is added. Otherwise exactly one sentence below is appended. This applies to correct and incorrect final answers.

C-MISSING-TRIGGER | H01 not asked, but H02 or H03 asked

For a similar presentation, asking whether standing as well as walking brings on symptoms can help establish the trigger pattern.

C-MISSING-RELIEF | H01 asked, but neither H02 nor H03 asked

For a similar presentation, asking how symptoms respond to sitting and lumbar flexion can help distinguish the leading possibilities.

C-MISSING-BOTH | H01 not asked and neither H02 nor H03 asked

For a similar presentation, asking about symptom triggers and relief with sitting or lumbar flexion can help establish the key clinical pattern.

10 / Incorrect final diagnosis and submission feedback

C-INCORRECT-HIGH | High confidence

Your final diagnosis differs from the most likely explanation here, despite your high confidence. This is a useful opportunity to revisit which findings favoured your choice and which pointed elsewhere. In this case, relief with sitting and forward flexion favours a neurogenic cause. Compare these features with the alternatives below to guide your next assessment.

C-INCORRECT-MODERATE | Moderate confidence

Your final diagnosis differs from the most likely explanation here. Use the comparisons below to strengthen your next assessment, especially how standing, sitting, and forward flexion affect symptoms.

C-INCORRECT-LOW | Low confidence

It is reasonable to feel uncertain when several diagnoses can cause leg discomfort. Your final diagnosis differs from the most likely explanation here, but the findings below offer a clear way forward. Focus on how sitting and spinal posture help distinguish the alternatives.

When the key inquiry combination is incomplete, append the applicable sentence from section 09. No numerical confidence score is calculated.

F01 | Incorrect first attempt - case remains active

Not quite. One diagnostic attempt remains. You can investigate further and revise your differential before trying again.

F02 | Debrief is available

Your debrief is available below.

F03 | Missing input - attempt not consumed

Enter your final diagnosis before submitting.

Select a confidence rating from 1 to 5 before submitting.

11 / Accepted final diagnoses and leading terms

Complete current list of 48 phrases. This same list produces Leading diagnosis in the ranked differential. Entries are matched after case, accent, punctuation and whitespace normalization. There is no fuzzy matching, substring matching, or interpretation of arbitrary sentences.

1. neurogenic claudication	25. narrowing of the lumbar spinal canal
2. lumbar spinal stenosis	26. neurogenic intermittent claudication due to lumbar stenosis
3. lumbar stenosis	27. intermittent neurogenic claudication
4. spinal stenosis	28. pseudoclaudication
5. lumbar canal stenosis	29. pseudo claudication
6. degenerative lumbar spinal stenosis	30. spinal claudication
7. spinal stenosis with neurogenic claudication	31. intermittent spinal claudication
8. neurogenic intermittent claudication	32. lumbar spinal stenosis resulting in neurogenic claudication
9. lumbar spinal stenosis causing neurogenic claudication	33. lumbar stenosis with neurogenic claudication
10. LSS	34. lumbar stenosis causing neurogenic claudication
11. lumbar spinal canal stenosis	35. lumbar stenosis resulting in neurogenic claudication
12. lumbar central canal stenosis	36. lumbar canal stenosis with neurogenic claudication
13. central lumbar spinal stenosis	37. lumbar canal stenosis causing neurogenic claudication
14. degenerative lumbar stenosis	38. lumbar canal stenosis resulting in neurogenic claudication
15. lumbar spinal stenosis with neurogenic claudication	39. degenerative lumbar spinal stenosis with neurogenic claudication
16. neurogenic claudication due to lumbar spinal stenosis	40. degenerative lumbar spinal stenosis causing neurogenic claudication
17. neurogenic claudication secondary to lumbar spinal stenosis	41. degenerative lumbar spinal stenosis resulting in neurogenic claudication
18. lumbar central stenosis	42. neurogenic claudication from lumbar spinal stenosis
19. degenerative lumbar canal stenosis	43. neurogenic claudication due to lumbar stenosis
20. degenerative lumbar spinal canal stenosis	44. neurogenic claudication secondary to lumbar stenosis
21. lumbar spine stenosis	45. neurogenic claudication from lumbar stenosis
22. stenosis of the lumbar spine	46. neurogenic claudication due to lumbar canal stenosis
23. stenosis of the lumbar spinal canal	47. neurogenic claudication secondary to lumbar canal stenosis
24. lumbar spinal canal narrowing	48. neurogenic claudication from lumbar canal stenosis

A final answer must match a listed diagnosis after normalization. A relevant alternative is not accepted as this case's correct final answer. Vague terms such as claudication alone or stenosis alone are not recognized as the intended diagnosis.

12 / Recognized relevant alternatives

Complete current list of 57 phrases. These receive Relevant alternative in the learner's differential, not a correct final-answer result. The list covers related diagnostic families; membership does not imply that every term has an identical clinical definition.

1. peripheral arterial disease	30. lumbar radicular syndrome
2. peripheral artery disease	31. lumbosacral radicular syndrome
3. PAD	32. radicular pain / radiculopathy
4. lower extremity peripheral arterial disease	33. lumbar radicular pain / radiculopathy
5. lower extremity peripheral artery disease	34. lumbar radiculitis
6. vascular claudication	35. lumbosacral radiculitis
7. arterial claudication	36. nerve root pain
8. intermittent vascular claudication	37. neuropathy
9. peripheral arterial disease / vascular claudication	38. distal peripheral neuropathy
10. lumbar radicular pain	39. distal polyneuropathy
11. bilateral lumbar radicular pain	40. distal symmetrical polyneuropathy
12. lumbosacral radicular pain	41. length dependent neuropathy
13. radicular leg pain	42. length dependent polyneuropathy
14. lumbar nerve root pain	43. peripheral sensory neuropathy
15. sciatica	44. distal sensory polyneuropathy
16. lumbar radiculopathy	45. lower limb peripheral neuropathy
17. lumbosacral radiculopathy	46. lower extremity peripheral neuropathy
18. peripheral neuropathy	47. lower limb peripheral arterial disease
19. peripheral polyneuropathy	48. lower limb peripheral artery disease
20. polyneuropathy	49. atherosclerotic peripheral arterial disease
21. distal symmetric polyneuropathy	50. atherosclerotic peripheral artery disease
22. length dependent peripheral neuropathy	51. intermittent arterial claudication
23. diabetic peripheral neuropathy	52. arterial intermittent claudication
24. diabetic neuropathy	53. PAD with claudication
25. radicular pain	54. peripheral arterial disease with claudication
26. radiculopathy	55. peripheral artery disease with claudication
27. bilateral radicular pain	56. peripheral artery disease / vascular claudication
28. bilateral radiculopathy	57. vascular intermittent claudication
29. bilateral lumbar radiculopathy	

Unlisted entries remain Not assessed. The game preserves their wording and does not infer that they are clinically inappropriate. It also does not automatically recognize negated, speculative or compound answers unless the complete normalized phrase is listed.

13 / Shared interface wording and states

These are shared controls and interface messages rather than additional clinical content. Their wording is included here so future cases can retain the same overall game.

U01 | Clinical Inquiry

Choose findings that help distinguish your leading possibilities.

U02 | Working Differential

Optional: add up to three diagnoses. Rank them using the arrows or drag to reorder.

U03 | Patient Record

Newest findings appear first.

U04 | Examination initially locked

Ask at least one history question to unlock the examination.

U05 | Examination unlocked

Examination available.

U06 | Initial attempt message

2 diagnostic attempts remaining. Submit whenever you are ready.

U07 | After first attempt

1 diagnostic attempt remaining. Submit whenever you are ready.

U08 | Case ended

Diagnostic attempts complete.

U09 | Controls and field wording

History | Examination | Imaging | Final Diagnosis | Submit Diagnosis | How confident are you? | Very uncertain | Very confident | Add | Restart | Restart challenge | In progress | Complete | All Challenges

Input placeholders: Add a possible diagnosis / Your most likely diagnosis. Both text inputs allow up to 250 characters. The initial referral entry is labelled Initial referral information.

U10 | Working differential validation

Enter a possible diagnosis first.

Remove an entry before adding another diagnosis.

That diagnosis is already in your differential.

U11 | Loading and unavailable content

Loading challenge...

Enable JavaScript to play this challenge.

This challenge could not be loaded. Return to All Challenges and try the available challenge again.

Future cases without imaging options can display: No previous imaging information is available for this patient.

14 / Brief for an additional case

Use the existing case as a model and supply the clinical content below. The two-attempt workflow, patient record, optional ranked differential, confidence selector, and responsive layout can remain the same.

NEW-01 | Case identity and initial presentation

Give a short title, difficulty level, patient vignette and intended working diagnosis. Avoid giving the diagnosis away in the starting information.

NEW-02 | Inquiry options and exact responses

For each History, Examination and Imaging inquiry, provide a compact button label and the exact response to add to the Patient Record. State explicitly when imaging has not been done.

NEW-03 | Acceptable diagnostic wording

List genuinely acceptable final-answer equivalents and abbreviations. List the relevant alternative families and their recognized terms separately. Specify broad or misleading terms that should not be accepted.

NEW-04 | Diagnosis and alternatives

Provide a concise explanation of the intended diagnosis. For each important alternative, explain why it is plausible and what distinguishes it. Include any dangerous alternative that needs a Must not miss discussion.

NEW-05 | Findings and evidence threshold

Choose the primary discriminating inquiries and write an Asked explanation and a Not asked explanation for each. Choose supporting findings and their priority. Identify the inquiry combination that establishes the key clinical pattern.

NEW-06 | Practical next steps

Provide concise assessment, investigation and management entries appropriate to the information available. Include relevant urgent warning symptoms without inventing missing clinical details.

NEW-07 | Personalized feedback and reading

Supply or adapt correct and incorrect feedback for Low, Moderate and High confidence, plus brief advice when key inquiries were not asked. Provide verified clinical reading links and short descriptions.

NOTES FOR YOUR NEXT CASE / REQUESTED CHANGES